

④ Statistics:- deals with collection, organisation & Analysis, interpretation & presentation of Data.

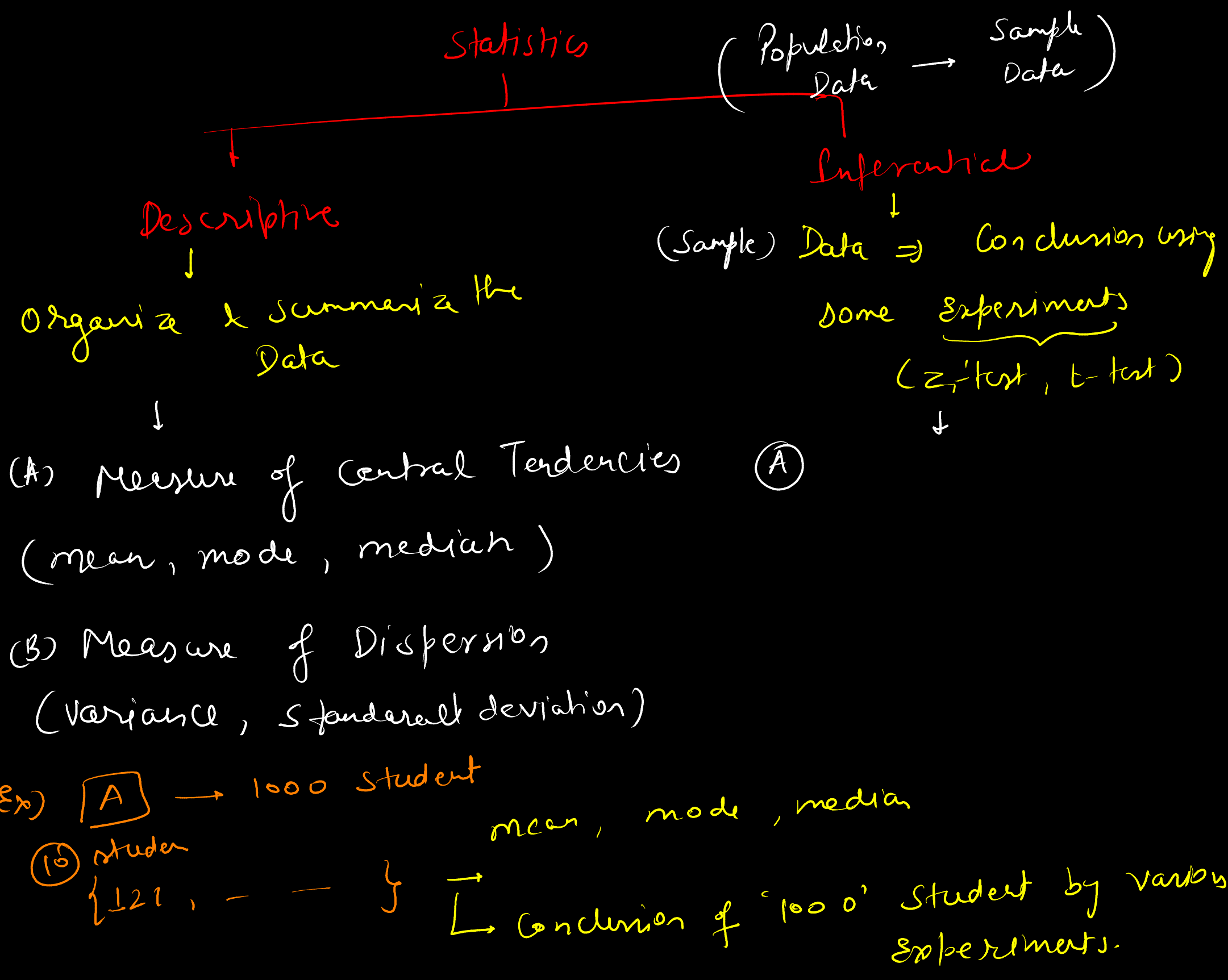
Data → organized (way) → prediction  
↓  
(statistical tools)

→ Statistical tools used to understand the data & predict something based on Data. (statistical Decision)

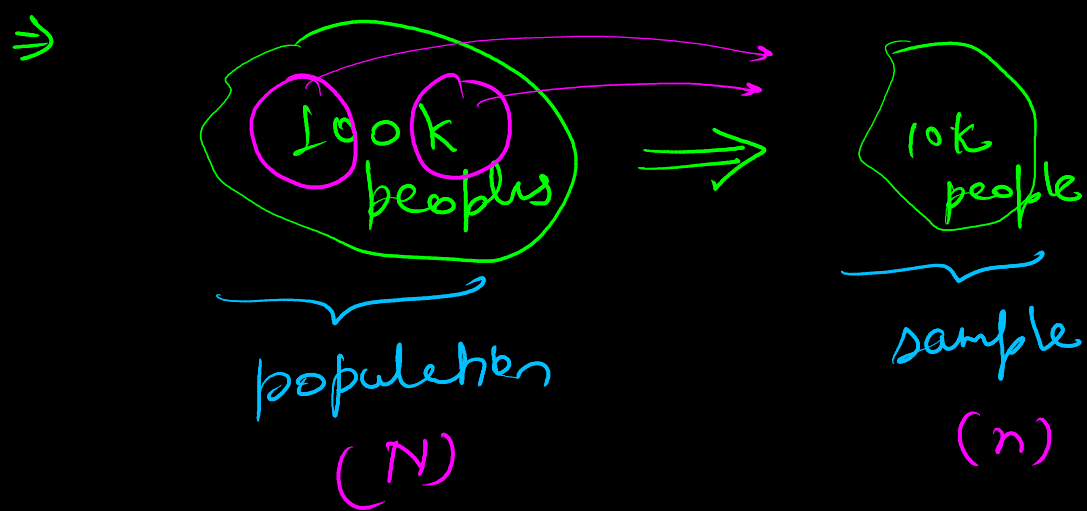
- ① Mean
- ② Median
- ③ Distributions
- ④ max, min, sum
- ⑤ charts, Graph
- ⑥ Pdf, cdf
- ;

⇒ Applications:-

↳ Machine Learning & Data Science  
↳ Data Analyst



# \* Population vs Sample Data :-



(10):- weight of all the peoples?

[Ex] → Exit poll

## \* Measure of central tendencies :-

① Mean (Average)

$N$  (population)

$n$  (size)

② Median

①

$$\mu_N = \frac{\sum x_i}{N}$$

$$\mu_n = \frac{\sum x_i}{n}$$

} mean

③ Mode

②

Sort & pick (central element)

$A = \{1, 3, 4, 5\}$   $\xrightarrow{\text{outlier}}$   $A = \{1, 3, 4, 5, 100\}$

mean  
 $\mu = \frac{1+3+4+5}{4} = \frac{13}{4} = 3.25$   $\xrightarrow{\text{outlier}}$   $\mu = 22.6$

median

Sort - Central element

$A = \{1, 3, 4, 5\}$

$A = \{1, 3, 4, 5, 100\}$

$(3, 4) \rightarrow \frac{3+4}{2} = \frac{7}{2} = 3.5$

[mode] - most Repeated element in Data

Missing values

mode Imputation

mean Imputation

median Imputation

# ⑧ Measure of Dispersion (against mean)

## ① Variance & Standard Deviation

$$\text{Age 1} = \{2, 2, 4, 4\}$$

$$n = 4$$

$$\text{var} = (2-3)^2 + (2-3)^2 + (4-3)^2 + (4-3)^2$$

$$\boxed{\text{var} = \sum (x_i - \mu)^2}$$

Check

$$\boxed{\sigma^2 =}$$
 → Variance

$$\boxed{\sigma = \sqrt{}}$$
 → S.D

$$\text{Age 2} = \{1, 1, 5, 5\}$$

$$n = 4$$

$$\text{var} = (1-3)^2 + (1-5)^2 + (5-3)^2 + (5-3)^2$$

$$= 4 + 4 + 4 + 4$$

$$= 16$$

$$\boxed{\sigma^2 = 16}$$

$$\boxed{\sigma = \sqrt{16}}$$

for sample size

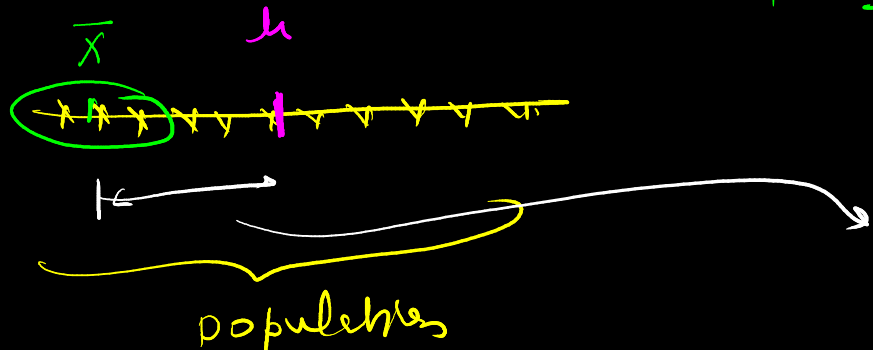
$$\boxed{\sigma_s^2 = \sum_{i=1}^n \frac{(x_i - \bar{x})^2}{n-1}}$$

for population size

$$\boxed{\sigma_p^2 = \sum_{i=1}^N \frac{(x_i - \mu)^2}{N}}$$

(Bessel Correction)

⇒ why  $n-1$ ?? —



$$s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$$

(underestimating the true population variance.)

(To reduce the mean sq. error)

(\*) Population (N)

Sample (n)

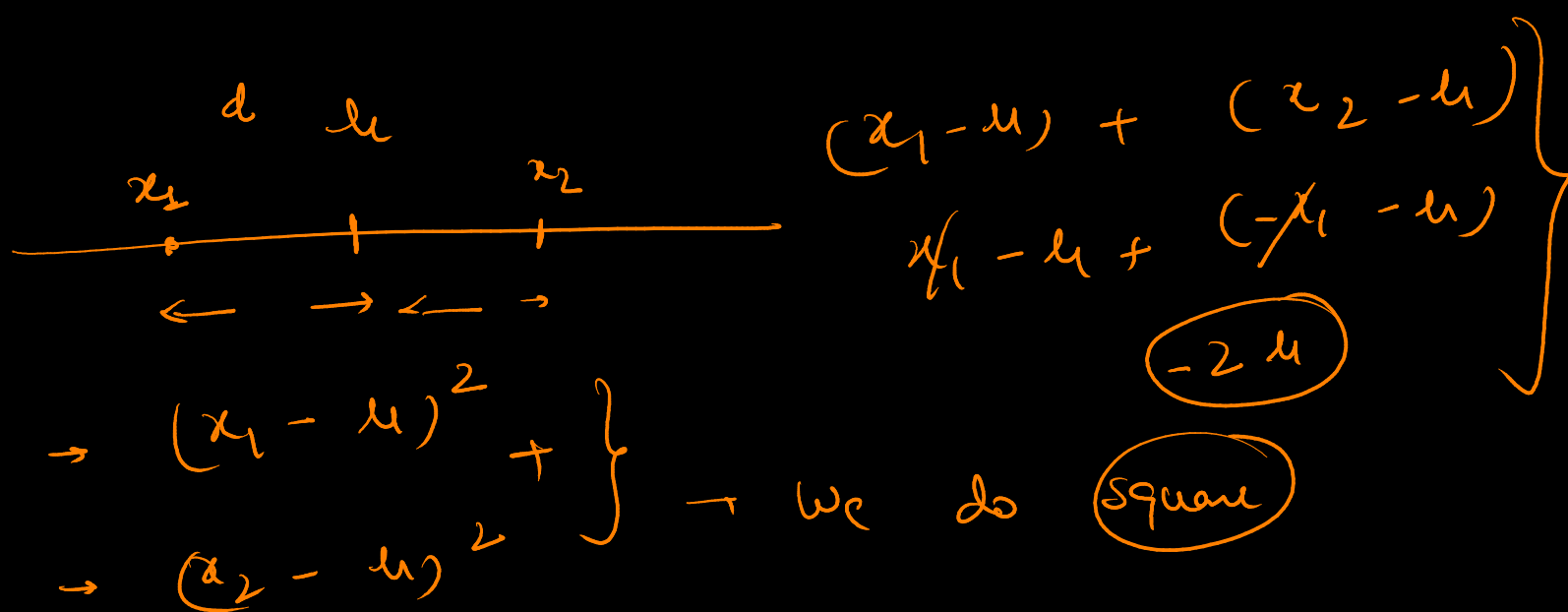
$$\mu = \frac{\sum (x_i)}{N}$$

$$\bar{x} = \frac{\sum (x_i)}{n}$$

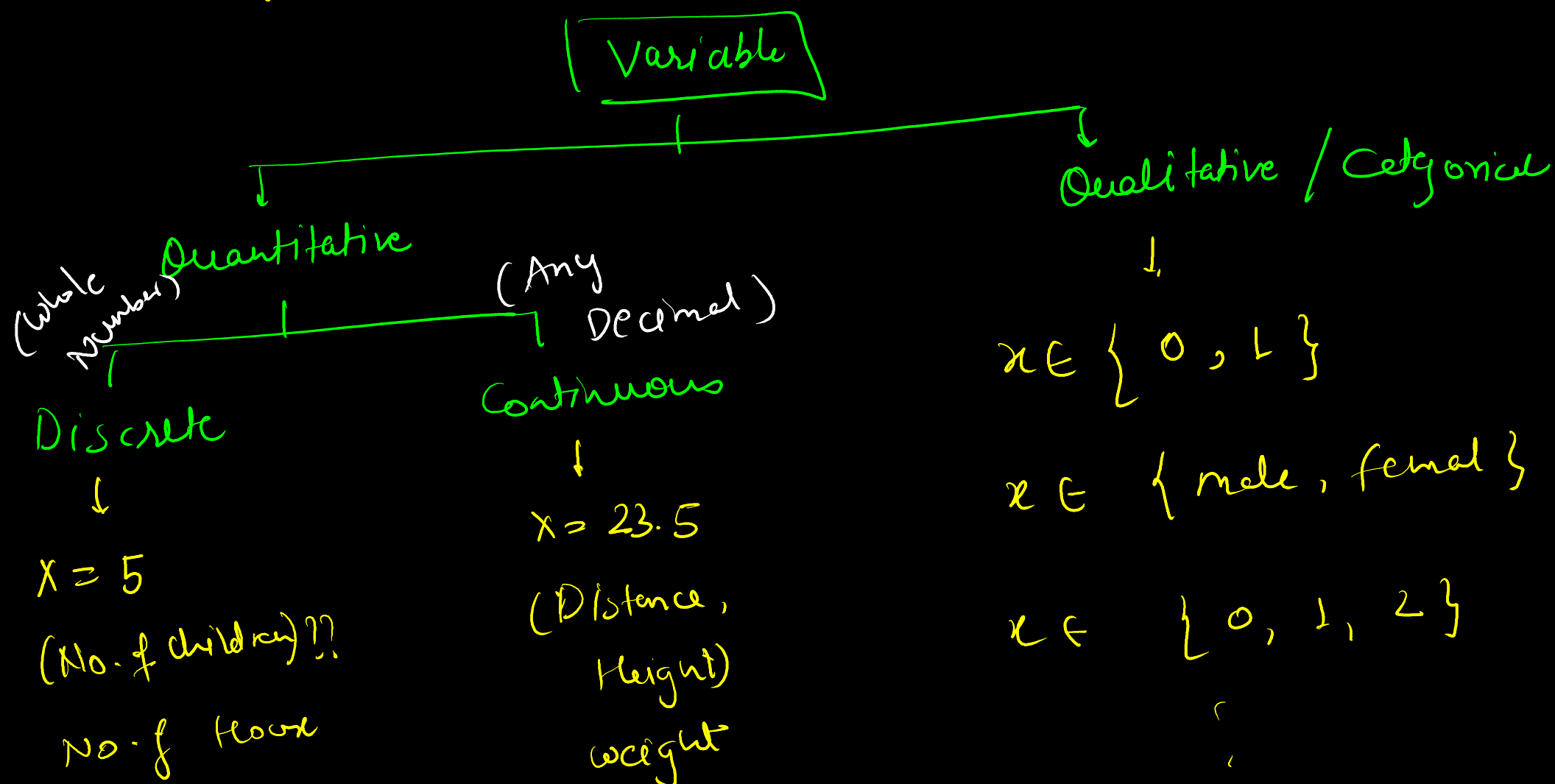
$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

SD =  $\sqrt{\sigma^2} = \sigma$  → How far is the Data point away from mean



(\*) Variables: which can take up some value  
Age = 32      gender = 'male'



## ④ Random variables :-

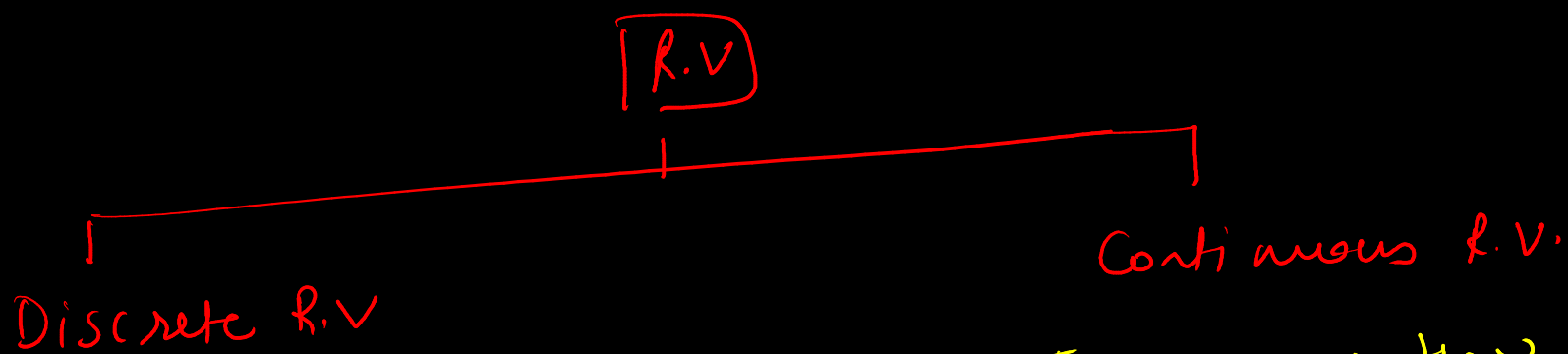
$X =$  is a  $f^h$  whose value are derived from diff. Experi

### ④ Tossing <sup>(Two)</sup> ~~a~~ Coin

$X =$  no. of Head  $= \{0, 1, 2\}$

### ④ Rolling of a fair Dice

$X =$  No. at top of ~~face~~  $= \{1, 2, 3, 4, 5, 6\}$



## ⇒ Histograms :-

$X = \{23, 24, 25, 30, 34, 36, 40, 45, 50, 60, 75, 80\}$

20 to 30  $\Rightarrow$  4 points

30 to 40  $\Rightarrow$  3

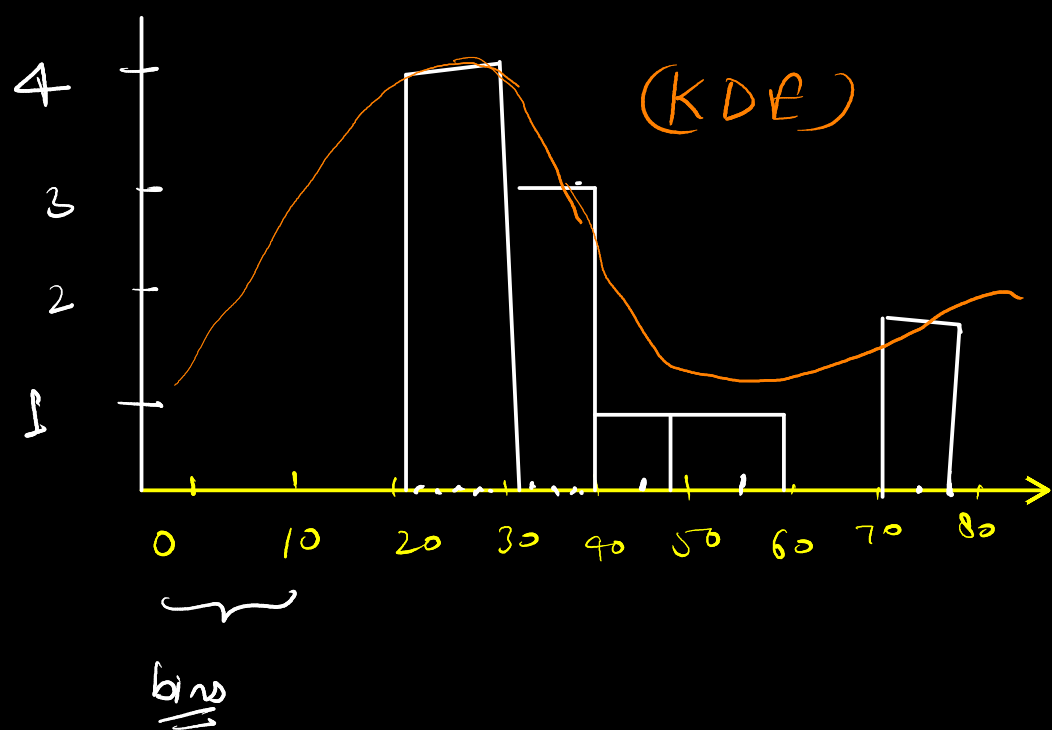
40 to 50  $\Rightarrow$  1

50 to 60  $\Rightarrow$  2

60 to 70  $\Rightarrow$  0

70 to 80  $\Rightarrow$  2

Count  
↓  
Ranging b/w  
bins



Kernel Density Estimator  $\Rightarrow$  Draw a smoother curve based on Histogram  
for Pdf

## ⇒ Percentiles & Quantiles :-

→ Percentage :-

percentage of odd No

$$X = \{1, 2, \check{3}, 4, \check{5}, 6\}$$

$$= 3/6 = \underline{\underline{50\%}}$$

→ Percentiles :-

↳ It is a value below which certain percentage of observation lies.

{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 8, 9, 9, 10}

$$\text{percentile of } 9 = \frac{\# \text{ of value below } 9}{n} \times 100 = \frac{11}{14} \times 100 = \underline{\underline{78\%}}$$

$$\text{percentile of } (\hat{x}) = \frac{\# \text{ values below } x}{n} \times 100$$

78% of entire distribution is less than  $(\hat{x})$   
↓  
[9]

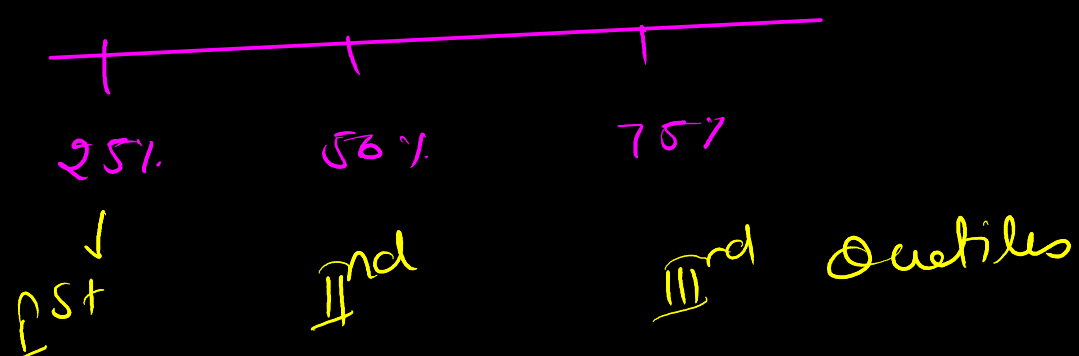
⇒ Q:- 25% of entire distribution ⇒  $\left( \frac{\text{Percentile} \times (n+1)}{100} \right)$

$$\downarrow \quad \frac{25}{100} = \frac{x}{(n+1)} \Rightarrow x = \frac{25}{100} \times (14+1) = 3.75 \approx \frac{3+4}{2} = \underline{\underline{3.5}}$$

(25%)

② 25% of entire distribution less than (3.5)

⇒ Quantiles :- Quantiles are value which divide the dataset into four equal part, Each contain 25% of Data.





⇒ 5 Number Summary :-  $D = \{1, 2, 2^2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 27\}$   
 $n = 19$

① Minimum

② 1<sup>st</sup> Quantile (25%)

③ Median

④ 3<sup>rd</sup> Quantile (75%)

⑤ Maximum

① Remove outliers

[lower fence value → higher fence value]

$$\text{lower fence value} = Q_1 - \underbrace{(IQR) * 1.5}_{\text{Inter Quartile Range (Q3 - Q1)}}$$

first quartile

$$\text{higher fence value} = Q_3 + (1.5) * IQR$$

②

$$Q_1 = \frac{25}{100} \times (n+1) = \frac{25}{100} \times 20 = 5 \Rightarrow \boxed{3}$$

(position)

↳ 25% of entire distribution is less than 3

$$Q_3 = \frac{75}{100} \times 20 = 15 \Rightarrow \boxed{7}$$

↳ 75% of entire distribution is less than 7

Range: [lower fence value ↔ higher fence value]

$$Lfv = Q_1 - (1.5) * (IQR)$$

$$= 3 - (1.5) * (4)$$

$$= 3 - 6 \Rightarrow \boxed{-3}$$

$$hfv = Q_3 + (1.5) * IQR$$

$$= 7 + 6 = \boxed{13}$$

$[-3, 13] \Rightarrow$  outside this range is (outlier)

So, 27 is outlier

(new)

$D = \{1, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 6, 7, 8, 8, 9, 21\}$   
 $n = 18$

min = 1

max = 9

median =  $\frac{5+5}{2} = 5$

$$Q1 = \frac{25}{100} * (19) = 4.75 \Rightarrow \frac{4+5}{2} = 4.5 \quad \frac{2+3}{2} = 2.5 \approx 3$$

$$Q3 = \frac{75}{100} * (19) = 14.25 \Rightarrow$$

Box Plot — to visualize the Outliers

\* Covariance & Correlation :-

⇒ used to determine the Relationship b/w two variable

① Covariance :- Covariance is measured of how much two random variable change together.

→ If the variable tends to increase & decrease together the Covariance is (+ve)  $A \propto B$  (directly proportional)

→ If the variable tend to increase & other decrease then the Covariance is (-ve)  $(A \propto \frac{1}{B})$  (indirectly proportional)

$$\text{Cov}(X, Y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n-1}$$

→ Disadvantage

$$-\infty < \text{Cov}(X, Y) < \infty$$

$$\text{Cov}(X, X) = \text{var}(X)$$

we need to limit

→ It help to Quantify Relationship b/w variable (Random)

$$[-1 \text{ to } 1]$$



⇒ Correlation:-

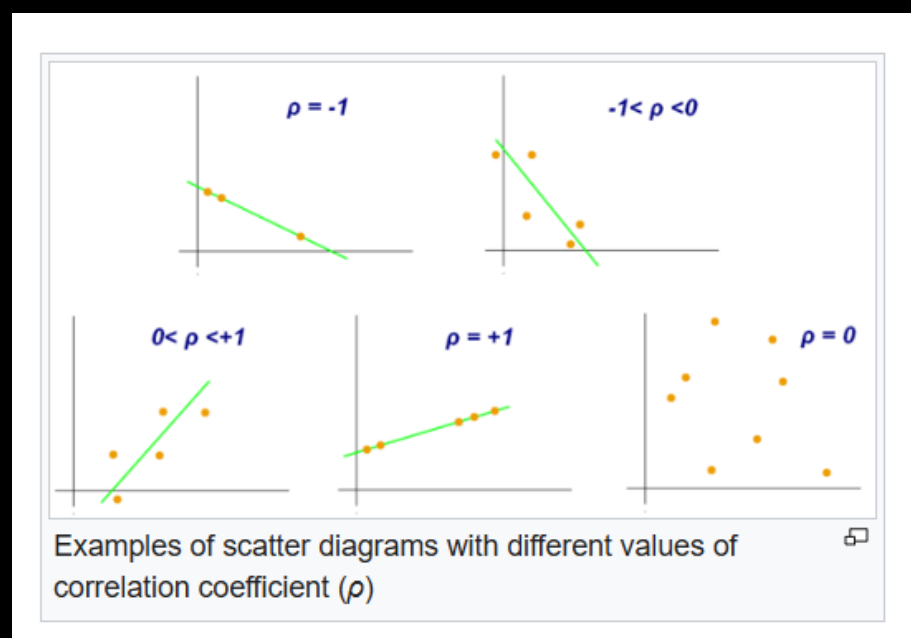
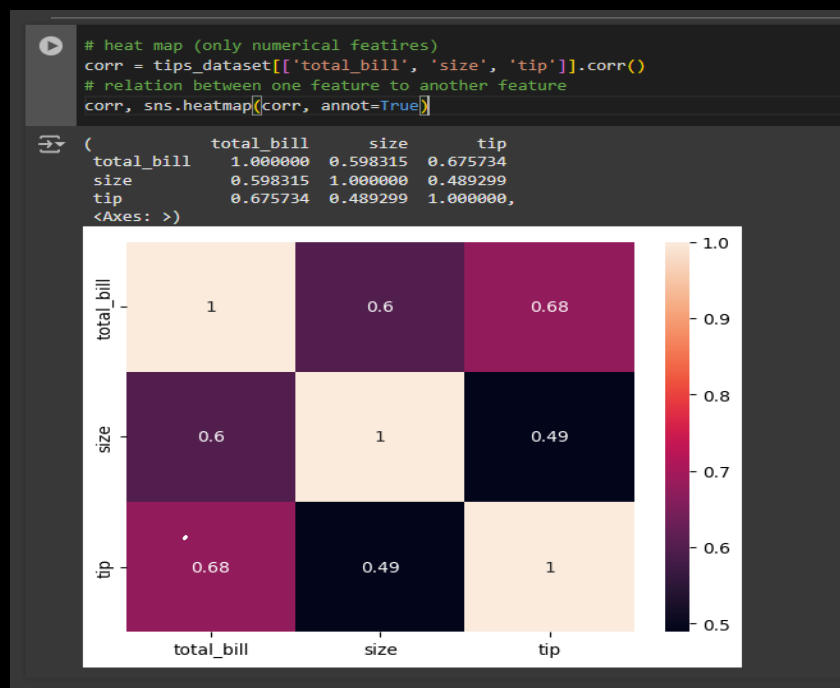
- ↳ Pearson Correlation Coefficient
- ↳ Spearman Rank Coefficient

① Pearson Correlation Coefficient :-  $[-1, 1]$

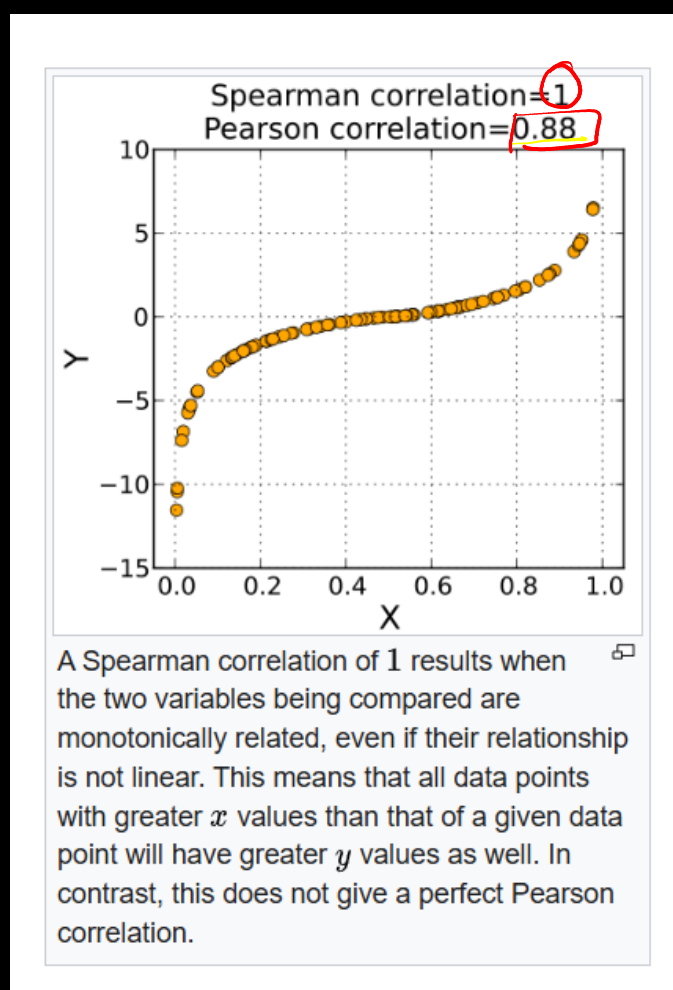
$$\rho_{x,y} = \frac{\text{Cov}(x,y)}{\sigma_x \sigma_y}$$

→ more the value towards  $(+1)$  the more +ve correlated

→ more the value towards  $(-1)$  then more -ve correlated



② Spearman Rank Correlation:-



→ Only linear Data will get

① by Pearson Correlation

$x \uparrow y \uparrow$

$x \downarrow y \downarrow$

~~~~~

Pearson Coefficient

$$= 0.88$$

→ for non-linear Data get some value

⇒ Pearson Correlation Coefficient does not Capture the non-linear Data.

$$\rho_s = \frac{\text{Cov}(R(x) - R(y))}{\sigma(R(x)) * \sigma(R(y))} \quad R(x) \text{ rank}$$

| <b>IQ, <math>X_i</math></b> | <b>Hours of TV per week, <math>Y_i</math></b> | <b>rank <math>x_i</math></b> | <b>rank <math>y_i</math></b> | <b><math>d_i</math></b> | <b><math>d_i^2</math></b> |
|-----------------------------|-----------------------------------------------|------------------------------|------------------------------|-------------------------|---------------------------|
| 86                          | 2                                             | 1                            | 1                            | 0                       | 0                         |
| 97                          | 20                                            | 2                            | 6                            | -4                      | 16                        |
| 99                          | 28                                            | 3                            | 8                            | -5                      | 25                        |
| 100                         | 27                                            | 4                            | 7                            | -3                      | 9                         |
| 101                         | 50                                            | 5                            | 10                           | -5                      | 25                        |
| 103                         | 29                                            | 6                            | 9                            | -3                      | 9                         |
| 106                         | 7                                             | 7                            | 3                            | 4                       | 16                        |
| 110                         | 17                                            | 8                            | 5                            | 3                       | 9                         |
| 112                         | 6                                             | 9                            | 2                            | 7                       | 49                        |
| 113                         | 12                                            | 10                           | 4                            | 6                       | 36                        |

✳️ Can be used in feature selection ✓

✳️ If feature are not correlated Can be Drop from the data

$\approx 0$

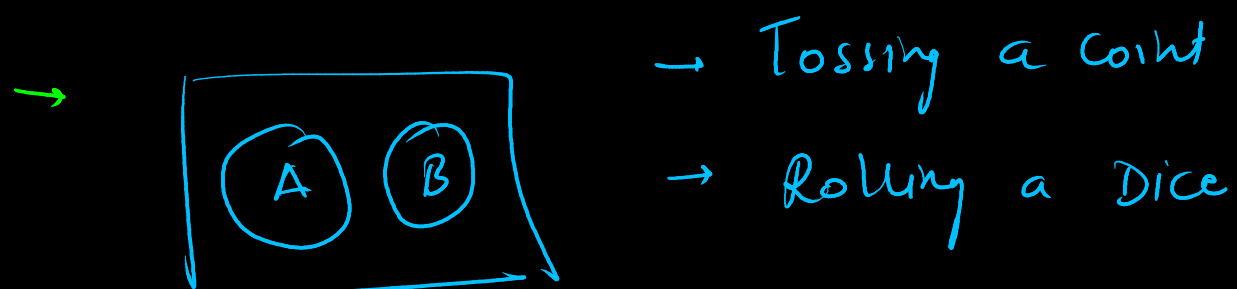
## ⑧ Probability :-

### ↳ Addition & Multiplication Rule

Probability :- Determining the likelihood of an experiment

Ex  $P_r(H) = \frac{1}{2}$   $P_r(T) = \frac{1}{2}$   $P_r(\text{getting 1 on dice}) = \frac{1}{6}$

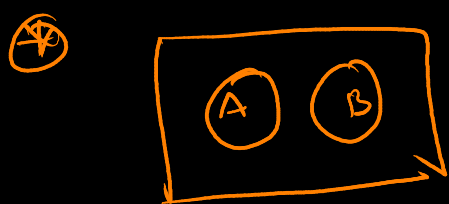
Mutually Exclusive Event :- If two or more event can't occur at same time ( $A \cap B = \emptyset$ )



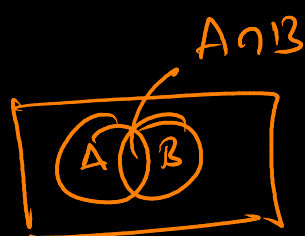
$$P\{A \cap B\} = 0$$

$$P(A \cup B) = P(A) + P(B) \quad \text{— mutually exclusive Event}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad \text{— Not mutual Exclusive Event}$$



mutually exclusive Event



Not mutual  
Exclusive Event

## ⑨ Multiplication Rule :-

Independent events :- If they don't effect one another

Ex - Tossing a coin Two time ( $H \rightarrow T$ )  
 $P(H) = \frac{1}{2}$   $P(T) = \frac{1}{2}$  (No change)

Dependent event :- If they effect each other  
one event  $\rightarrow$  other event

①  $\begin{bmatrix} 2R \\ 3B \end{bmatrix} \rightarrow 1R \text{ ball}$   
 $P(\text{draw } R) = \frac{2}{5} \Rightarrow \begin{bmatrix} 1R \\ 3B \end{bmatrix} \Rightarrow P(B) = \frac{3}{4}$

③ Independent Event Sample space changes

$$P(A \cup B) = P(A) \cdot P(B) \quad \text{--- (Independent Event)}$$

$$P(A \cup B) \rightarrow P(A) \cdot P(B/A) \quad \text{--- (Dependent Event)}$$

↓  
Cond<sup>n</sup> probability

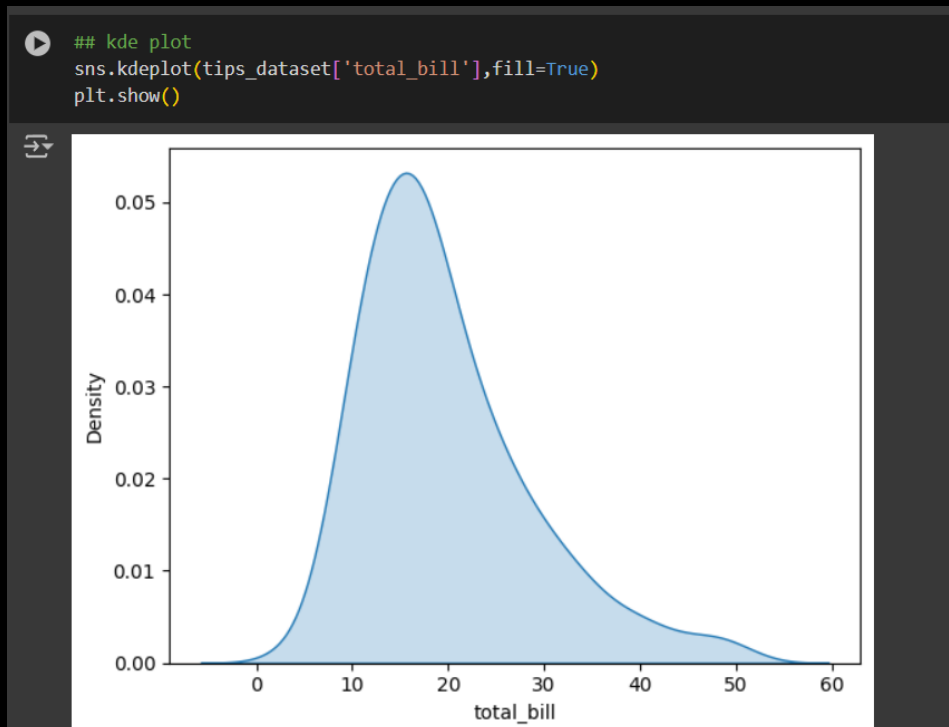
\* Probability Distribution function :- describe how the probabilities are distributed over the values of Random variable

Pdf

↓

① Probability mass function (pmf)  
↳ Discrete R.V

② Probability Density f<sup>n</sup> (pdf)  
↳ Continuous R.V



\* Probability Mass function (PMF)

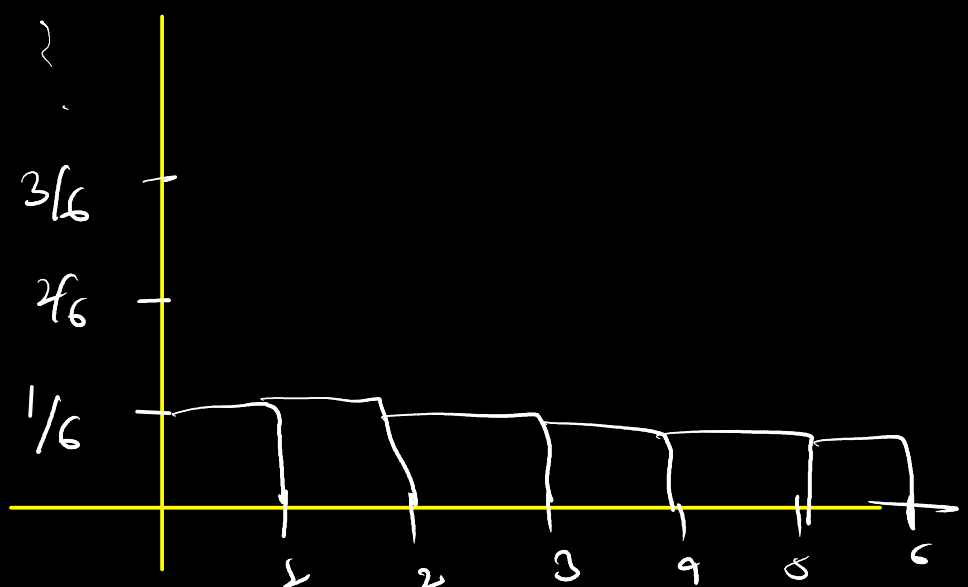
↳ Based on Discrete R.V

Ex-1) Rolling a fair dice

$X = \{ \text{getting a no. on dice (face)} \}$

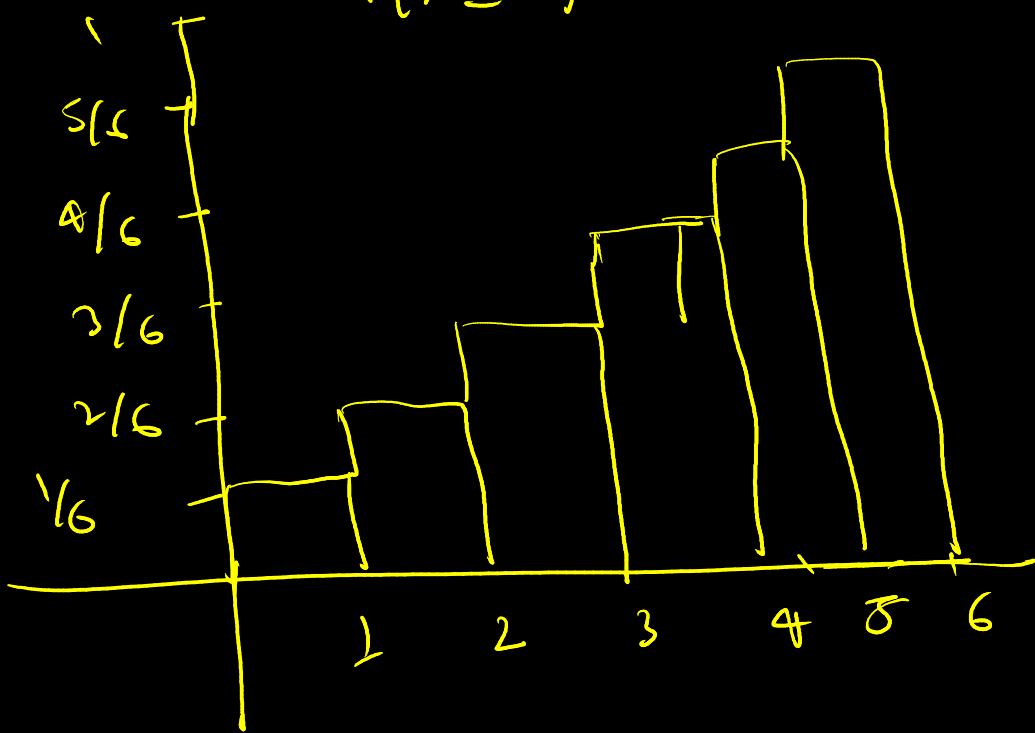
$$P[X=x] = 1/6$$

| X        | 1     | 2     | 3     | 4     | 5     | 6     |
|----------|-------|-------|-------|-------|-------|-------|
| $P[X=x]$ | $1/6$ | $1/6$ | $1/6$ | $1/6$ | $1/6$ | $1/6$ |



# Cdf (Cumulative Density function)

|               |     |       |       |       |       |       |       |
|---------------|-----|-------|-------|-------|-------|-------|-------|
| $P[X \leq x]$ | $x$ | 1     | 2     | 3     | 4     | 5     | 6     |
| $P[X \leq x]$ |     | $1/6$ | $2/6$ | $3/6$ | $4/6$ | $5/6$ | $6/6$ |



PMF  $\Rightarrow P[X=x]$

Cdf  $\Rightarrow P[X \leq x]$

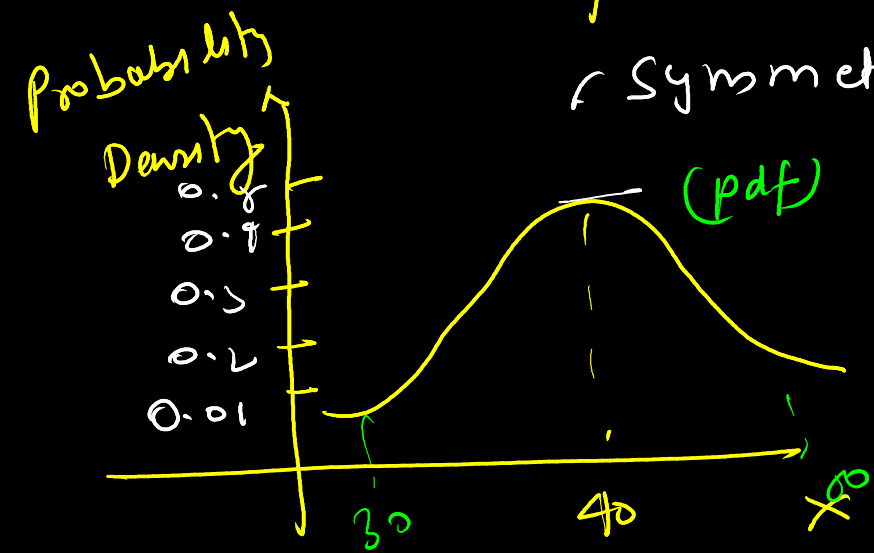
## \* Probability Density function

↳ for continuous variable (Kde plot)

↳ Histogram  $\rightarrow$  Smoother it

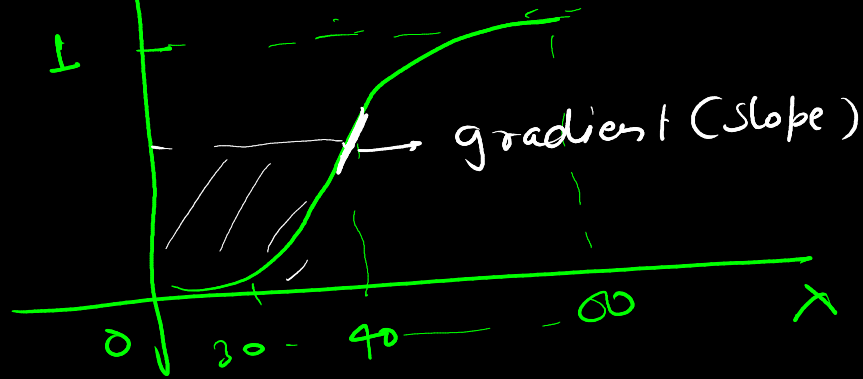
↳ Symmetric (let say)

① Area under the curve



cdf  $\Rightarrow$

Cumulative prob



$P[X=x]$

$P[X \leq x]$

$$P[X \leq x] = \int_a^b f(x) \cdot dx \quad \text{--- cdf}$$

\*  
//

Probability Density  $\Rightarrow$  Gradient of Cumulative density function

Properties of Pdf  $\Rightarrow$  (increasing)

① Always non-negative  $f(x) \geq 0 \forall x$

② the total area under curve of Pdf is ①

$$\int_{-\infty}^{\infty} f(x) \cdot dx = 1$$

⊕ Types of Distribution :-

→ Each Dataset Can follow some kind of Distributions.

(i) Bernoulli Distribution  $\rightarrow (n=1)$

↳ outcome is Binary  
↳ Discrete R.V

(ii) Binomial Distribution -  $(n > 1)$

↳ Discrete R.V (pmf)

(iii) Normal / Gaussian Distribution

↳ Continuous R.V (pdf)

(iv) Poisson's Distribution

↳ Continuous R.V (pdf)

(v) Log normal Distribution (pdf)

(vi) Uniform normal Distribution (pmf)

Helps in (EDA & FE) as Data Analyst



# ⇒ Bernoulli Distribution :-

0 → Success  
1 → failure



The Bernoulli distribution is one of the simplest and most fundamental probability distributions in statistics and probability theory. It models a random experiment that has exactly two possible outcomes: "success" (typically coded as 1) and "failure" (typically coded as 0).

## Key Characteristics

1. **Definition:** A random variable  $X$  follows a Bernoulli distribution if:

$$X = \{0, 1\}$$

$$P(X = 1) = p \quad \text{and} \quad P(X = 0) = 1 - p$$

where  $p$  is the probability of success ( $0 \leq p \leq 1$ ).

2. **Probability Mass Function (PMF):** The PMF of  $X$  is:

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

3. **Parameters:**

✓  $p$ : Probability of success.

4. **Mean (Expected Value):**

$$E[X] = p$$

5. **Variance:**

$$\text{Var}(X) = p(1 - p)$$

6. **Support:**  $X$  takes values in  $\{0, 1\}$ .

## Applications

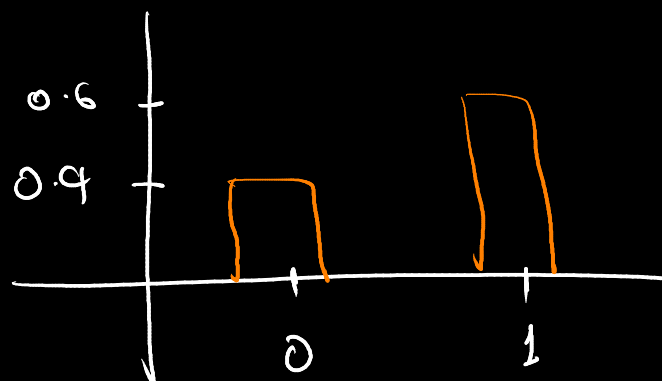
- Representing binary outcomes like coin flips, yes/no answers, or success/failure in experiments.
- Building blocks of the Binomial distribution (sum of  $n$  independent Bernoulli trials).

(pmf) — Company has launched smartphone 'A'

used — 60% ⇒  $p = 0.6$

not used — 40% ⇒  $p = 0.4$

$$P(X = x) = p^x (1 - p)^{1-x} \quad (\text{PMF})$$



$$P(X = 1) \Rightarrow p, \quad P(X = 0) = (1 - p) = (0.4)$$

$$\text{pmf} = \begin{cases} 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \end{cases}$$

④  $E(X) = p \Rightarrow$  (mean)

$$E(X) = \sum_{i=0}^1 x \cdot p(X=x) = 0 \cdot p(X=0) + 1 \cdot p(X=1)$$

$$\boxed{E(X) = p}$$

④  $V(X) = p(1-p) \Rightarrow$  variance  $\Rightarrow p \cdot q$

$$V(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \sum_{i=0}^1 x^2 \cdot p(X=x) \Rightarrow 0^2 \cdot p(X=0) + 1^2 \cdot p(X=1)$$

$$\boxed{E(X^2) = p}$$

$$V(X) = p - (p^2) = p(1-p) = p \cdot q$$

④ Median (central element)

$$\text{Median} = \begin{cases} 0 & \text{if } p < 1/2 \\ [0, 1] & \text{if } p = 1/2 \\ 1 & \text{if } p > 1/2 \end{cases}$$

$$\text{median} = \begin{cases} 0 & q > p \\ & \text{threshold} \\ \boxed{0.5} & q = p \\ 1 & q < p \end{cases}$$

④ mode

$$p > q \Rightarrow \text{mode} = p$$

$$\text{else} \Rightarrow \text{mode} = (1-p)$$

④ Standard Deviation

$$\sigma = \sqrt{\sigma^2} = \sqrt{p(1-p)} = \sqrt{p \cdot q}$$



## ⇒ Binomial Distribution :-

The Binomial distribution is an extension of the Bernoulli distribution that models the number of successes in a fixed number of independent Bernoulli trials.

### Key Characteristics

1. **Definition:** A random variable  $X$  follows a Binomial distribution if:

$$X \sim \text{Binomial}(n, p)$$

where:

$$x \in \{0, 1\}$$

- $n$ : Number of trials.
  - $p$ : Probability of success in each trial.
2. **Probability Mass Function (PMF):** The probability of observing exactly  $k$  successes in  $n$  trials is given by:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, 2, \dots, n$$

where:

$k = \text{no. of successes}$

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Binomial  
Coefficient

is the binomial coefficient.

3. **Mean (Expected Value):**

$$\mathbb{E}[X] = n \cdot p$$

4. **Variance:**

$$\text{Var}(X) = n \cdot p \cdot (1 - p)$$

5. **Support:**  $X$  takes values in  $\{0, 1, 2, \dots, n\}$ .

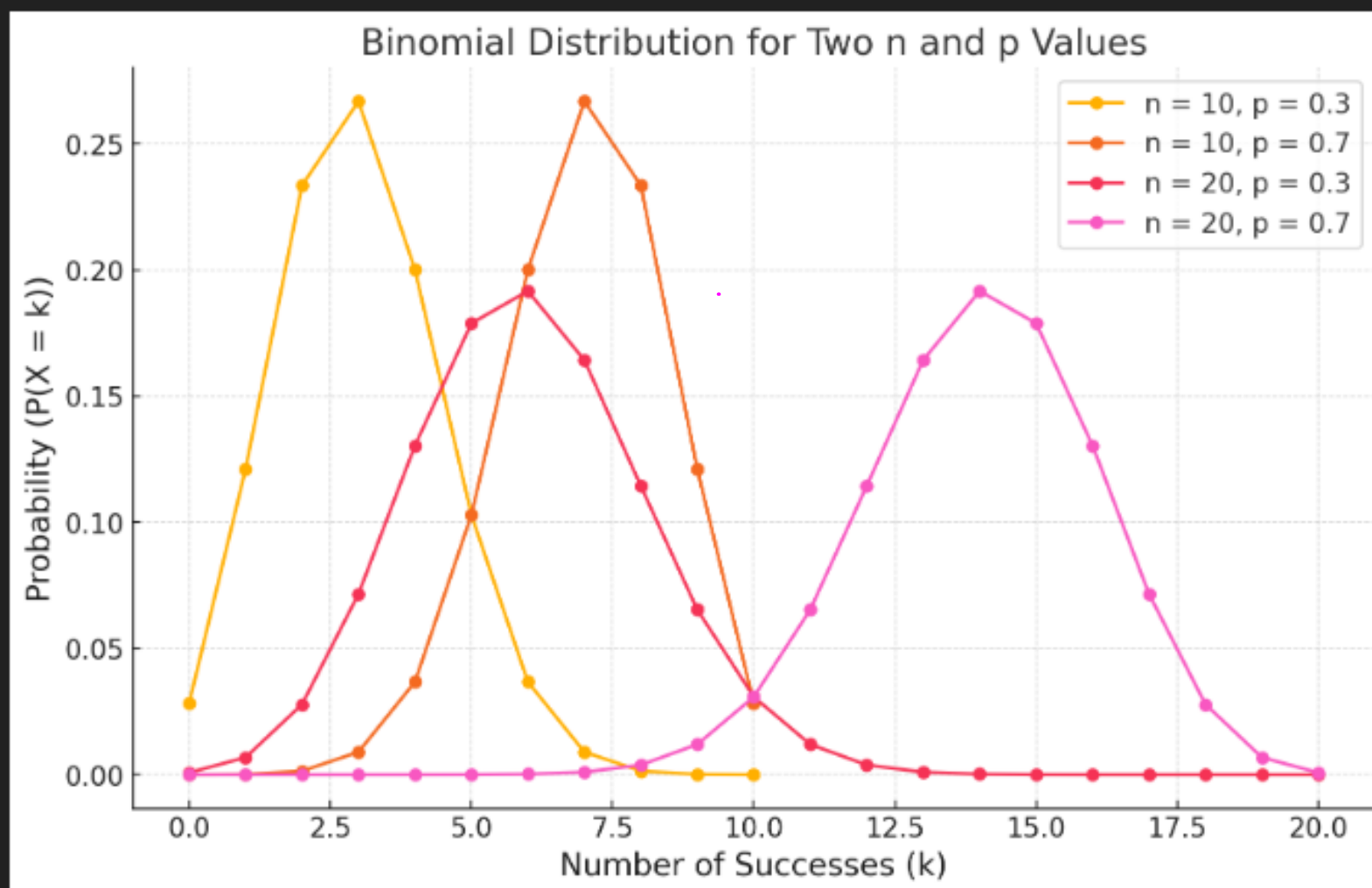
### Applications

- Counting the number of heads in  $n$  coin tosses.
- Modeling the number of defective items in a batch of  $n$  items.
- Estimating the number of correct answers in a multiple-choice test when guessing.

Q:) coin flip  $n = 5$

$$p = 0.5$$

$$\begin{aligned} P(\text{getting exactly } 3 \text{ heads}) &= \binom{n}{k} p^k (1-p)^{n-k} \\ &= \binom{5}{3} (0.5)^3 (0.5)^2 \end{aligned}$$



Here's the updated graph showing the Binomial distribution for two specific combinations of  $n$  and  $p$ :

- $n = 10, p = 0.3$ : The peak is closer to fewer successes, as the probability of success is smaller.
- $n = 10, p = 0.7$ : The peak shifts towards more successes due to a higher probability.
- $n = 20, p = 0.3$  and  $n = 20, p = 0.7$ : The distributions spread out more because the number of trials ( $n$ ) is larger.

Ex - Quality Control

$$\boxed{n=10} \Rightarrow p(\text{defective}) = 10\% = \underline{\underline{0.1}}$$

$$P(\text{exactly 2 defective item}) = {}^{10}C_2 (0.1)^2 (0.9)^8$$

⇒ Poisson's Distribution



The **Poisson distribution** is a discrete probability distribution that models the number of events occurring in a fixed interval of time or space, under the assumption that these events occur independently and at a constant average rate.

→ Discrete R.V (pmf)

→ No. of event occurring in fixed time interval  
Ex- No. of people visiting Bank (7 to 10) hrs

### Key Characteristics

1. **Definition:** A random variable  $X$  follows a Poisson distribution if:

$$X \sim \text{Poisson}(\lambda)$$

where:

- $\lambda > 0$ : The average number of events in the given interval (also called the rate parameter).

2. **Probability Mass Function (PMF):** The probability of observing exactly  $k$  events is given by:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

3. **Mean (Expected Value):**

$$\mathbb{E}[X] = \lambda$$

4. **Variance:**

$$\text{Var}(X) = \lambda$$

5. **Support:**  $X$  takes values in  $\{0, 1, 2, \dots\}$ .

### Assumptions

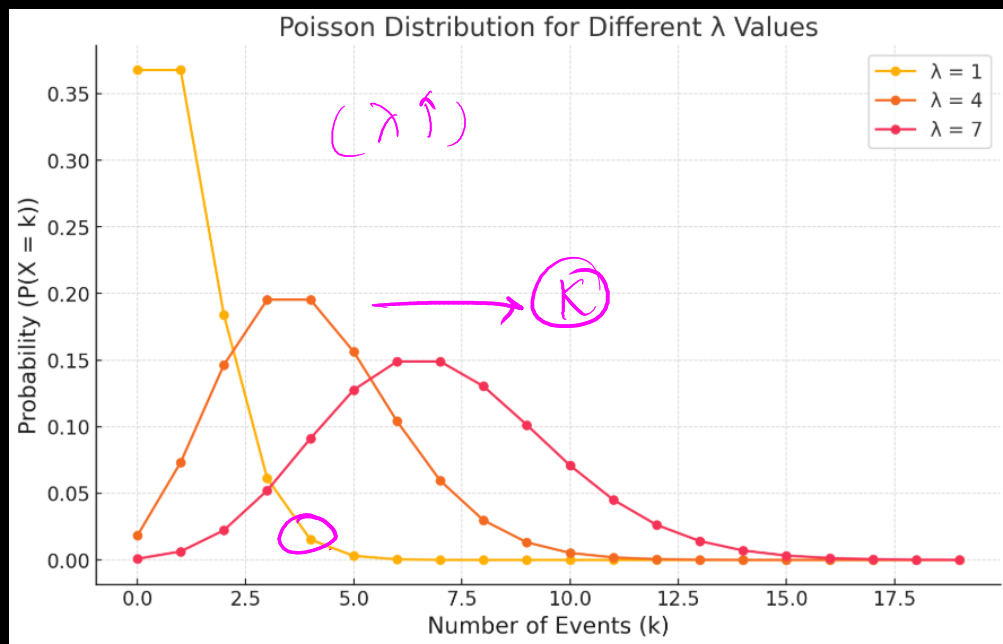
- Events occur **independently** of each other.
- The average rate ( $\lambda$ ) is constant over time or space.
- Two events cannot occur simultaneously (they are considered rare).

### Applications

- ✓ Modeling the number of customer arrivals at a service desk in an hour.
- ✓ Counting the number of phone calls received by a call center in a day.
- ✓ Predicting the number of decay events from a radioactive source in a fixed period.

## Relationship to Other Distributions

1. **Binomial to Poisson:** When  $n \rightarrow \infty$  and  $p \rightarrow 0$ , with  $n \cdot p = \lambda$ , the Binomial distribution approaches the Poisson distribution.
2. **Exponential Distribution:** If the time between events follows an exponential distribution, the number of events in a fixed interval follows a Poisson distribution.



Here's a graph showing the Poisson distribution for different values of  $\lambda$  (1, 4, and 7).

- For smaller  $\lambda$ , the distribution is concentrated near 0. ←
- As  $\lambda$  increases, the distribution spreads out and its peak shifts to higher values of  $k$ . ←

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import poisson

# Parameters for the Poisson distribution
lambdas = [1, 4, 7] # Different rate parameters (λ)
k_values = np.arange(0, 20) # Values of k

# Plotting
plt.figure(figsize=(10, 6))

for lam in lambdas:
    pmf_values = poisson.pmf(k_values, lam)
    plt.plot(k_values, pmf_values, marker='o', label=f'λ = {lam}')

# Adding labels, legend, and grid
plt.title('Poisson Distribution for Different λ Values', fontsize=16)
plt.xlabel('Number of Events (k)', fontsize=14)
plt.ylabel('Probability (P(X = k))', fontsize=14)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.grid(alpha=0.5)
plt.legend(fontsize=12)
```

⇒ Normal Gaussian Distribution :- (Continuous R.V.)  
(Bell curve)

```
python

from scipy.stats import norm

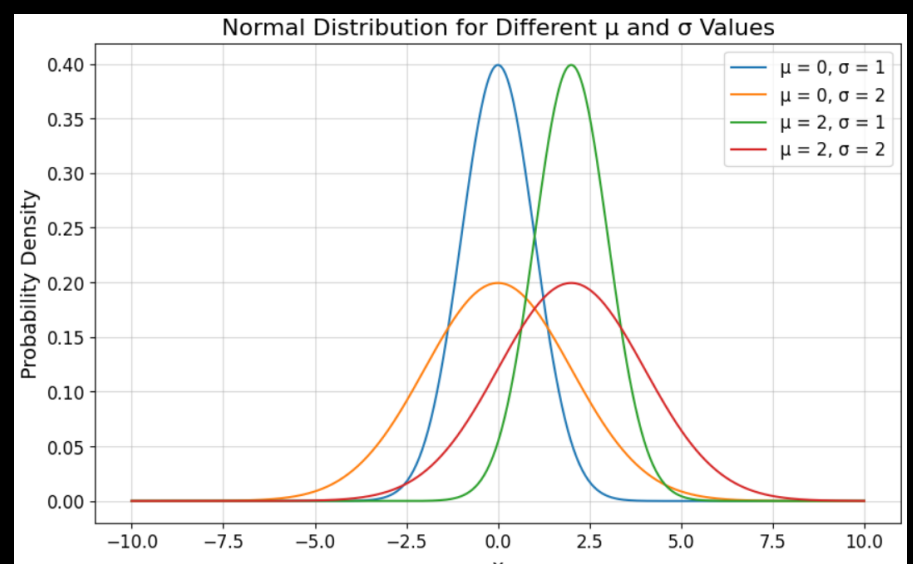
# Parameters for the Normal (Gaussian) distribution
means = [0, 2] # Different means (μ)
std_devs = [1, 2] # Different standard deviations (σ)

# Range for the x-axis
x = np.linspace(-10, 10, 500)

# Plotting
plt.figure(figsize=(10, 6))

for mean in means:
    for std_dev in std_devs:
        pdf_values = norm.pdf(x, mean, std_dev)
        plt.plot(x, pdf_values, label=f'μ = {mean}, σ = {std_dev}')

# Adding labels, legend, and grid
plt.title('Normal Distribution for Different μ and σ Values', fontsize=16)
plt.xlabel('x', fontsize=14)
plt.ylabel('Probability Density', fontsize=14)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.grid(alpha=0.5)
plt.legend(fontsize=12)
plt.show()
```



### Graph Behavior for Different Parameters:

1. **Changing  $\mu$ :**
  - Shifts the curve horizontally along the  $x$ -axis.
  - The peak of the curve is always at  $\mu$ .
2. **Changing  $\sigma$ :**
  - A smaller  $\sigma$  makes the curve narrower and taller (less spread).
  - A larger  $\sigma$  makes the curve wider and shorter (more spread).

→ weights of student in a class

LRCS Dataset

The **Normal distribution**, also known as the **Gaussian distribution**, is a continuous probability distribution widely used in statistics, data science, and natural sciences. It is characterized by its symmetric, bell-shaped curve and is often used to model real-world phenomena.

Key Characteristics

1. **Definition:** A random variable  $X$  follows a Normal distribution if:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

where:

- $\mu$ : Mean (center of the distribution).
- $\sigma^2$ : Variance (measure of spread).

2. **Probability Density Function (PDF):** The PDF of  $X$  is given by:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- $\mu$ : Determines the location of the peak.
- $\sigma$ : Determines the spread or width of the curve.

3. **Mean, Median, and Mode:** For a Normal distribution, these are all equal and located at  $\mu$ .

4. **Variance and Standard Deviation:**

- Variance:  $\sigma^2$
- Standard Deviation:  $\sigma$

5. **Symmetry:** The distribution is symmetric about the mean ( $\mu$ ).

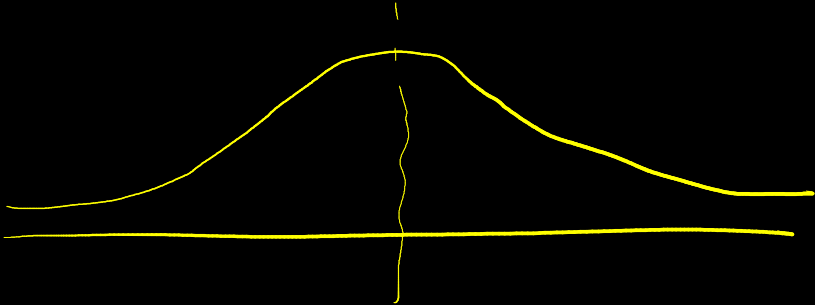
6. **68-95-99.7 Rule (Empirical Rule):**

- 68% of data falls within 1 standard deviation of the mean.
- 95% falls within 2 standard deviations.
- 99.7% falls within 3 standard deviations.

Applications

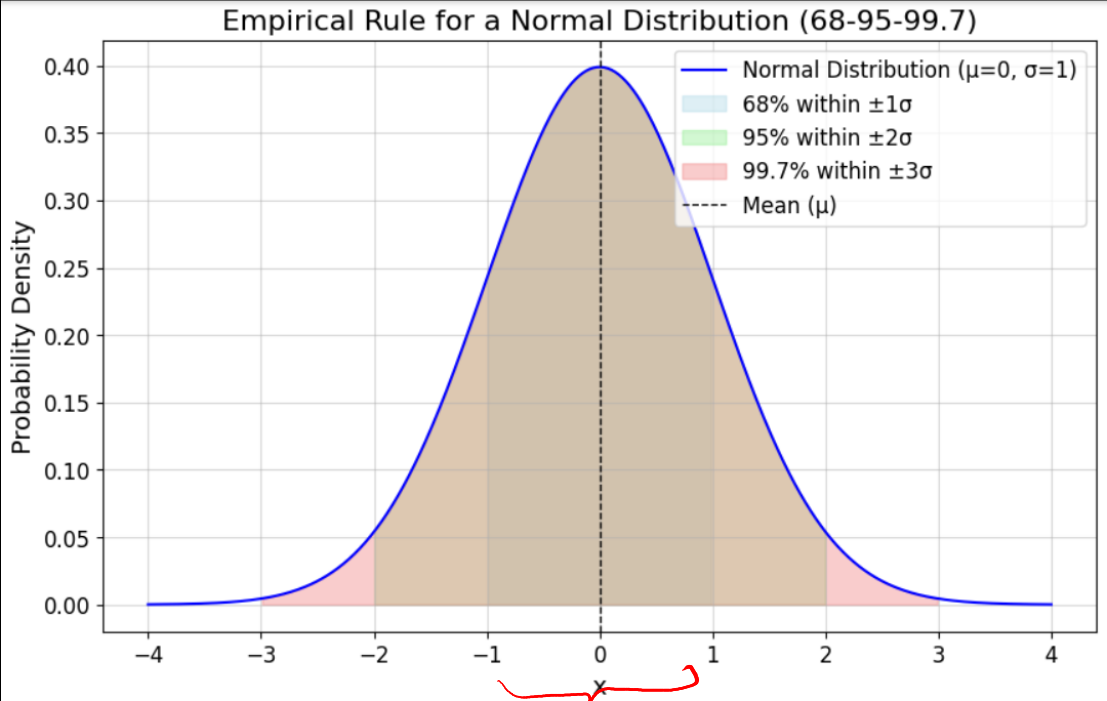
- Heights, weights, and test scores in a population.
- Errors in measurement or forecasting.
- Financial models (e.g., stock prices approximations).

→ Mean = median = mode  
- Symmetric



$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\mu = \frac{\sum x_i}{N} \qquad \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$



68%  
95%  
99.7%

```
# Plotting the Empirical Rule (68-95-99.7) for a Normal Distribution
mean = 0 # Mean (μ)
std_dev = 1 # Standard deviation (σ)

# Range for the x-axis
x = np.linspace(-4, 4, 500)
pdf_values = norm.pdf(x, mean, std_dev)

# Plot the normal curve
plt.figure(figsize=(10, 6))
plt.plot(x, pdf_values, label='Normal Distribution (μ=0, σ=1)', color='blue')

# Shade regions for 68%, 95%, and 99.7%
regions = [1, 2, 3]
colors = ['lightblue', 'lightgreen', 'lightcoral']
labels = ['68% within ±1σ', '95% within ±2σ', '99.7% within ±3σ']

for i, region in enumerate(regions):
    plt.fill_between(x, 0, pdf_values, where=(x >= -region*std_dev) & (x <= region*std_dev),
                    color=colors[i], alpha=0.4, label=labels[i])

# Adding labels, legend, and grid
plt.title('Empirical Rule for a Normal Distribution (68-95-99.7)', fontsize=16)
plt.xlabel('x', fontsize=14)
plt.ylabel('Probability Density', fontsize=14)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)
plt.axvline(mean, color='black', linestyle='--', linewidth=1, label='Mean (μ)')
plt.grid(alpha=0.5)
plt.legend(fontsize=12)
plt.show()
```



The **standard normal distribution** is a special case of the normal distribution where the mean ( $\mu$ ) is 0 and the standard deviation ( $\sigma$ ) is 1. It is used to describe a normal distribution with these specific parameters and is represented by the **Z-distribution**. The shape of the distribution is symmetric and bell-shaped, and it is used in various statistical methods to standardize values.

## Z-Score

A **Z-score** represents the number of standard deviations a data point is from the mean of a distribution. It tells you how far away a specific value is from the mean in terms of standard deviations.

The formula to calculate a Z-score is:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- $X$  is the value of the data point.
- $\mu$  is the mean of the population or distribution.
- $\sigma$  is the standard deviation of the population or distribution.

A **positive Z-score** means the data point is above the mean, and a **negative Z-score** means the data point is below the mean.

## Using Z-scores

Z-scores are useful in:

1. **Comparing different distributions:** Since Z-scores standardize values, you can compare values from different distributions.
2. **Finding probabilities:** Using the standard normal distribution, you can look up the probability of a Z-score occurring in a standard normal table or use software to compute it.

Standard Normal Distribut  
( $\mu=0$   $\sigma^2=1$ )

$N \sim (\mu, \sigma^2)$

$\Downarrow$  using Z-score  
 $N_s \sim (0, 1)$

Standardization

Data  $\xrightarrow{\text{Z-score}}$  Standardize Data

SMD - all will get converted to same unit

$\Rightarrow$  Uniform Distribution

① Continuous UD (pdf)

② Discrete UD (pmf)

$\rightarrow$  Continuous UD:-

### Example:

If you have a uniform distribution between 1 and 5, the PDF is constant between 1 and 5, and the probability of any value occurring within this range is the same for all values. The mean would be  $\mu = \frac{1+5}{2} = 3$ , and the variance would be  $\sigma^2 = \frac{(5-1)^2}{12} = \frac{16}{12} = \frac{4}{3}$ .

### Applications:

Continuous uniform distributions are used in simulations, random number generation, and modeling scenarios where each outcome within a range is equally likely (e.g., randomly selecting a number from 1 to 10).

The **continuous uniform distribution** is a type of probability distribution where all outcomes are equally likely within a given range. This means that if you randomly select a value from the distribution, each possible value within the specified interval has the same probability of occurring.

### Characteristics of a Continuous Uniform Distribution:

- **Range:** The distribution is defined over a continuous interval, typically denoted as  $[a, b]$ , where  $a$  is the lower bound and  $b$  is the upper bound.
- **Probability Density Function (PDF):** The probability density function of a continuous uniform distribution is constant over the interval  $[a, b]$ . The formula for the PDF is:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

The height of the PDF is  $\frac{1}{b-a}$ , ensuring that the total area under the curve is equal to 1 (as required for any probability distribution).

- **Cumulative Distribution Function (CDF):** The cumulative distribution function (CDF) gives the probability that a random variable  $X$  is less than or equal to a particular value  $x$ . For a continuous uniform distribution, the CDF is:

$$F(x) = \begin{cases} 0 & \text{if } x < a \\ \frac{x-a}{b-a} & \text{if } a \leq x \leq b \\ 1 & \text{if } x > b \end{cases}$$

- **Mean and Variance:**

- The mean of a continuous uniform distribution is:  
$$\mu = \frac{a+b}{2}$$

- The variance is:

$$\sigma^2 = \frac{(b-a)^2}{12}$$

1. Probability Density Function (PDF)

The PDF for a continuous uniform distribution is constant within the interval  $[a, b]$  and zero outside this interval. It is given by:

f(x) = { 1/(b-a) if a <= x <= b, 0 otherwise }

- The height of the PDF is 1/(b-a), ensuring that the total probability under the curve is 1. This means that the probability is uniformly distributed over the interval.

2. Cumulative Distribution Function (CDF)

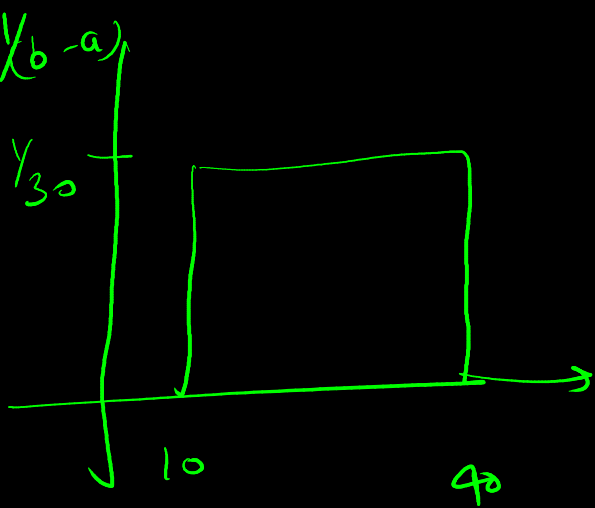
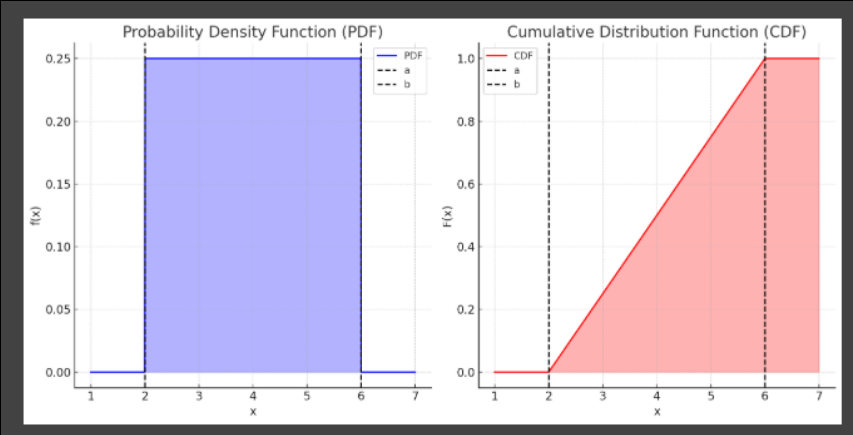
The CDF gives the probability that a random variable X takes a value less than or equal to a specific value x. The CDF for a continuous uniform distribution is:

F(x) = { 0 if x < a, (x-a)/(b-a) if a <= x <= b, 1 if x > b }

- For x < a, the probability is 0 (since x is outside the distribution's range).
- For a <= x <= b, the probability increases linearly from 0 to 1.
- For x > b, the probability is 1 (since x is greater than the upper bound of the distribution).

Summary of Key Points:

- The PDF is a constant value 1/(b-a) within the interval [a, b].
- The CDF increases linearly from 0 at x = a to 1 at x = b.
- The distribution is uniform, meaning each value within the interval has the same probability of being selected.



P(15 <= x <= 30) = (x2 - x1) \* 1/(b-a) = (30 - 15) \* 1/30 = 0.5

P(x >= 20) = (40 - 20) \* 1/30 = 2/3

=> Uniform Discrete Distribution

Definition of Uniform Discrete Distribution:

A uniform discrete distribution is a type of probability distribution where each possible outcome in a finite sample space has an equal chance of occurring. It is characterized by the fact that all outcomes have the same probability.

Mathematically, if a discrete random variable X takes values from a finite set {x1, x2, ..., xn}, then the probability mass function (PMF) of X is given by:

P(X = xi) = 1/n, for all i = 1, 2, ..., n

Where:

- n is the number of distinct outcomes,
- Each outcome has the same probability, which is 1/n.

Example:

Consider a die roll, where the die has 6 faces numbered from 1 to 6. The probability of each outcome is the same, i.e.,

P(X = 1) = P(X = 2) = P(X = 3) = P(X = 4) = P(X = 5) = P(X = 6) = 1/6

Graph:

For a uniform discrete distribution, the graph consists of vertical bars where the height of each bar is equal, reflecting the same probability for each outcome.

Let me generate a graph of a uniform discrete distribution for a die roll (6 outcomes).

